Exam Date & Time: 27-Nov-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

III SEMESTER B.TECH END SEMESTER EXAMINATIONS, NOV 2024

SIGNALS and SYSTEMS [BME 2125]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

Instructions to Candidates: Answer ALL questions. Missing data may be suitably assumed.

1) Using discrete-time Fourier transform, solve $y(n) = x(n) * h(n)$ where,

A) $x(n) = \{1, 1, 1, 1, -1, -1\}$ (3)
$\uparrow$

and

$h(n) = \delta(n-2) - \delta(n-4)$.

B) Interpret the statement - "The Z-transform is a more general version of DTFT." (2)

C) The input $x(n)$ and the impulse response $h(n)$ of a system is given by

$x(n) = u(n)$

&

$h(n) = \propto^n u(n)$ (5)

$0 < \propto < 1$

Compute the output $y(n)$ using Z-transform.

2) A) The ROCs of the Z-transforms of $x(n)$ and $x(n+n_0)$ are identical with the possible exception of $z = 0$ or $z = \infty$. Illustrate with suitable examples. (3)

B) Consider an LSI system described by the difference equation,

$$y(n) - \frac{1}{2}y(n-1) = x(n) + \frac{1}{2}x(n-1)$$ (3)

Determine the frequency response, $H(w)$, and impulse response, $h(n)$ of the system.

C) Analyze the discrete-time system, $y(n) = x(n) + x(n-1)$ and find out (with justification) whether the system is: (4)

(i) Linear (ii) Shift-invariant (iii) Causal (iv) Stable

3) The frequency response of the discrete-time low pass filter is given by, $H(w) = 1 + e^{-jw}$.

A) (i) Determine the impulse response $h(n)$ of the system. (3)

(ii)Determine and sketch the magnitude response $|H(w)|$ over $-\pi \leq w \leq +\pi$.

B) "The ROC of a left-sided sequence is the interior of a circle with radius R_{x+} i.e. $|Z| < R_{x+}$." Illustrate this with an example. (3)

C) Analyze the following linear-shift invariant system with the impulse response,

$$h(n) = \begin{cases} a^n, & n \geq 0 \\ b^n, & n < 0 \end{cases}$$

(4)

and identify (with justification) the range of the parameters $'a'$ and $'b'$ for which the system is:

(i) Stable (ii) Causal

4) Solve $x(n) * \delta(n - n_0)$ using both time-domain and frequency-domain methods. (4)

A)

B) Consider the interconnection of LSI systems, as shown in Fig. 1.

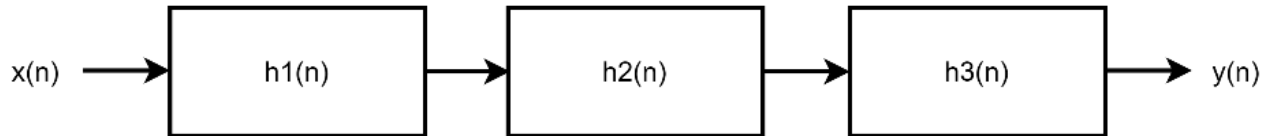


Fig. 1

Determine the overall impulse response in terms of $h1(n)$, $h2(n)$, and $h3(n)$ when (4)

$$h1(n) = \delta(n) - \delta(n-1)$$

$$h2(n) = u(n)$$

$$h3(n) = h(n)$$

C) Determine the response of the following system,

$$y(n) = x(2n + 2)$$

to the input signal $x(n)$ given below, using the precedence rule. (2)

$$x(n) = \begin{cases} |n|; & -3 \leq n \leq 3 \\ 0; & Otherwise \end{cases}$$

5) Consider a sequence, $x(n) = \left(-\frac{1}{2}\right)^n u(-n) + 2\left(\frac{1}{4}\right)^n u(n)$ (3)

A) (i) Indicate the characteristics of $x(n)$.

(ii) Based on the characteristics of $x(n)$, identify the ROC of the sequence without explicitly calculating the Z-transform.

B) Consider a three-point moving average filter,

$$y(n) = \frac{1}{3}[x(n+1) + x(n) + x(n-1)]$$

(i) Determine the system's impulse response. (4)

(ii) Determine the frequency response $H(w)$, and plot the magnitude of $H(w)$ for the system over $0 \leq w \leq +\pi$.

C) Analyze and find out (with justification) whether the LSI system,

$$h(n) = 2u(n) - 2u(n-1)$$

is both Causal and Stable. (3)

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